

```
!pip install pandas
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->panda
```

```
import pandas as pd
import numpy as np
```

```
#series
s=pd.Series([1,2,3,4,5,6,7])
print(s)
```

```
0    1
1    2
2    3
3    4
4    5
5    6
6    7
dtype: int64
```

```
#series with index
s=pd.Series([1,2,3,4,5,6,7],index=["a","b","c","d","e","f","g"])
print(s)
```

```
a    1
b    2
c    3
d    4
e    5
f    6
g    7
dtype: int64
```

```
print(s['b'])
```

```
2
```

```
d={'khasim':50,'farhan':60,'nawaz':70}
marks=pd.Series(d)
print(marks)
```

```
khasim    50
farhan    60
nawaz     70
dtype: int64
```

```
n=pd.Series([10,20,30,40,50])
n+2
```

```
   0
0  12
1  22
2  32
3  42
4  52
dtype: int64
```

```
n=pd.Series([10,20,30,40,50])
n.mean()
```

```
np.float64(30.0)
```

```
n.sum()
```

```
np.int64(150)
```

```
n.max()
```

```
50
```

```
data={
    'name':['khasim','nawaz','jabir','farhan','charan'],
    'age':[20,21,21,23,21],
    'salary':[2000,2000,3000,4000,3050],
    'city':['kadapa','hyderabad','goa','mumbai','kerala'],
    'gender':['male','male','male','male','male']
}
df=pd.DataFrame(data)
print(df)
```

	name	age	salary	city	gender
0	khasim	20	2000	kadapa	male
1	nawaz	21	2000	hyderabad	male
2	jabir	21	3000	goa	male
3	farhan	23	4000	mumbai	male
4	charan	21	3050	kerala	male

```
print(df.head(3))
```

	name	age	salary	city	gender
0	khasim	20	2000	kadapa	male
1	nawaz	20	2000	hyderabad	male
2	jabir	21	3000	goa	male

```
df.tail(3)
```

	name	age	salary	city	gender
2	jabir	21	3000	goa	male
3	farhan	22	4000	mumbai	male
4	charan	21	3050	kerala	male

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 5 columns):
#   Column  Non-Null Count  Dtype
---  ---
0   name    5 non-null        object
1   age     5 non-null        int64
2   salary  5 non-null        int64
3   city    5 non-null        object
4   gender  5 non-null        object
dtypes: int64(2), object(3)
memory usage: 332.0+ bytes
```

```
print(df.index)
```

```
RangeIndex(start=0, stop=5, step=1)
```

```
print(df.dtypes)
```

```
name      object
age        int64
salary    int64
```

```
city      object
gender    object
dtype: object
```

```
df.nunique()
```

```
      0
name  5
age   3
salary 4
city  5
gender 1

dtype: int64
```

```
df.isnull()
```

	name	age	salary	city	gender
0	False	False	False	False	False
1	False	True	False	False	False
2	False	False	False	False	False
3	False	True	False	False	False
4	False	False	False	False	False

```
df.isnull().sum()
```

```
      0
name  0
age   2
salary 0
city  0
gender 0

dtype: int64
```

```
df['name']
df.name
```

	name
0	khasim
1	nawaz
2	jabir
3	farhan
4	charan

dtype: object

```
df[['name', 'age']]
```

	name	age
0	khasim	20.0
1	nawaz	NaN
2	jabir	21.0
3	farhan	NaN
4	charan	21.0

```
df['age'].fillna(df['age'].mean(), inplace=True)
```

/tmp/ipykernel_1683/4255040007.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chainable assignment. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which w

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df

```
df['age'].fillna(df['age'].mean(), inplace=True)
```

```
df['age']
```

	age
0	20.000000
1	20.666667
2	21.000000
3	20.666667
4	21.000000

dtype: float64

```
df.drop(2)
```

	name	age	salary	city	gender
0	khasim	20.000000	2000	kadapa	male
1	nawaz	20.666667	2000	hyderabad	male
3	farhan	20.666667	4000	mumbai	male
4	charan	21.000000	3050	kerala	male

```
df[df['salary']>2000]
```

	name	age	salary	city	gender
2	jabir	21	3000	goa	male
3	farhan	23	4000	mumbai	male
4	charan	21	3050	kerala	male

```
def grade(salary):
    if salary >= 90:
        return 'A'
    elif salary >= 80:
        return 'B'
    else:
        return 'C'

# apply() with function
df['Grade_Letter'] = df['salary'].apply(grade)
```

```
print(df)
```

	name	age	salary	city	gender	Grade_Letter
0	khasim	20	2000	kadapa	male	A
1	nawaz	21	2000	hyderabad	male	A
2	jabir	21	3000	goa	male	A
3	farhan	23	4000	mumbai	male	A
4	charan	21	3050	kerala	male	A

```
df.groupby('salary').agg({
    'salary': ['mean', 'max', 'min']
})
```

salary			
	mean	max	min
salary			
2000	2000.0	2000	2000
3000	3000.0	3000	3000
3050	3050.0	3050	3050
4000	4000.0	4000	4000

```
df.groupby('salary').size()
```

0	
salary	
2000	2
3000	1
3050	1
4000	1

dtype: int64

```
employees_data = {
    'employee_id': [1, 2, 3, 4, 5],
    'name': ['khasim', 'nawaz', 'jabir', 'farhan', 'charan'],
    'dept_id': [101, 102, 101, 103, 102]
}
employees = pd.DataFrame(employees_data)

departments_data = {
    'dept_id': [101, 102, 103],
    'dept_name': ['Sales', 'Marketing', 'HR']
}
departments = pd.DataFrame(departments_data)

pd.merge(employees, departments, on='dept_id', how='left')
```

	employee_id	name	dept_id	dept_name
0	1	khasim	101	Sales
1	2	nawaz	102	Marketing
2	3	jabir	101	Sales
3	4	farhan	103	HR
4	5	charan	102	Marketing

